

H

Roll No.

TMC-401

M. C. A. (FOURTH SEMESTER) END SEMESTER

EXAMINATION, May, 2022

GRAPHICS AND VISUAL COMPUTING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among
(a), (b) and (c) in each main question.

(iii) Total marks in each main question are
twenty.

(iv) Each sub-question carries 10 marks.

1. (a) What is aliasing and anti-aliasing ? (CO2)
- (b) Calculate the pixel points along a straight line whose end points are A (30, 40) and B (50, 60). (CO2)
- (c) Discuss Bresenham ellipse drawing algorithm. (CO2)

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2. (a) Write mid-point circle algorithm and apply that algorithm to find the pixel values of the circle whose radius $r = 4$ and center of the circle is $(0, 0)$. (CO4)
- (b) Explain Cohen Sutherland Line clipping method with an example. (CO4)
- (c) Explain Weiler Atherton Polygon Clipping method. (CO4)
3. (a) Define Window and Viewport. Derive window to viewport transformation. (CO1)
- (b) What is meant by parallel and perspective projection ? Derive the matrix for perspective projection. (CO1)
- (c) What is the difference between geometric and coordinate transformations ? Discuss the various 2D geometric transformations. (CO1)
4. (a) Given control points $(10, 100)$, $(50, 100)$, $(70, 120)$, $(100, 150)$. Calculate coordinates of any *four* points lying on the corresponding B-spline curve. (CO3)

(3)

- (b) What is shading ? Discuss various forms of shading. (CO3)
 - (c) Describe the boundary fill algorithm with 4 connected pixels. Discuss with a suitable example. (CO3)
5. (a) What is Visualization and discuss its application ? (CO5)
- (b) Discuss Painter's algorithm. (CO5)
 - (c) Write short notes on the following : (CO5)
 - (i) Morphing
 - (ii) Shading
 - (iii) Fractal Graphics